

The Soyuz TMA-14 vehicle will remain docked to the ISS as an emergency escape pod, in case storm troopers barge in and launch a hostile takeover

The amount of platinum in an average 10-kilometre wide asteroid is worth \$5,000 billion. This value is not considering other minerals

not planned yet, another Shuttle mission has been sanctioned, and a space shuttle might be used for leftover operations on the ISS. Furthermore, sometime in 2010, the Chinese also plan to launch a space station of their own called the Tiangong 1. This will be done by a base launch of a module of eight tons, which will be joined by two or three *Shenzhou* spacecrafts over the span of a year to collectively form a small space station or research lab.

2011-2015

There are no firm plans yet. However, the Chinese are expected to accelerate their space activity in this period. The Tiangong 1 will be complemented with two more space stations, named the Tiangong 2 and Tiangong 3. This is important for the Chinese as a successful mission will give them a realistic idea of approaching manned missions to the moon. The approach for the Tiangong missions will be the same, which will involve a number of *Shenzhou* spacecraft launching to dock with the space stations. The Chinese will also launch an unmanned mission to Mars in 2014, as a pre-cursor to possible manned missions by 2040.

The Japanese Hayabusa 2 mission will be launched sometime in this period. This launch is significant as the spacecraft will test the possibility of human spaceflight to an asteroid, and back again. JAXA will also focus on its *Selene 2* project, which will orbit the moon and act as a precursor for manned missions to the moon around 2020. The Japanese manned flight program is lagging behind the Chinese and the Indians, but they are expected to be the fifth country to send humans into space.

ISRO is on schedule for being the fourth space agency, after RSA, NASA and CNSA, to send humans into space. The PSLV or GSLV rockets can be used as launch vehicles — the latter is a more likely choice. India has already demonstrated the capability of indigenous heat shields in a space capsule recovery experiment using PSLV rockets. A facility for launching the GSLV MK-II launch vehicle is expected to be constructed at the outskirts of Bengaluru around the year 2012. At the same time, training for human spaceflight will commence in India at Bengaluru. This facility is expected to involve large tanks of water for zero-gravity training. There will also be a centrifuge setup for training. The Indian version of the astronaut may be called a *gaganaut*, a *vishvanaut* or a *brahmanaut*.

The first of these will be on board a Russian Soyuz mission in 2013. The Russians will collaborate with the Indians on a number of issues, including selecting and training the crew, and technical help with life support systems. By 2015, the first two-man crew from India, on board an Indian spacecraft, using an Indian launch vehicle is expected to reach a low-earth orbit. This launch is expected to take off from the ISRO base at Sriharikota. The Orbital Vehicle used is going to be similar in design and function to the Soyuz. After spending about ten days in orbit, the space craft will return to the Bay of Bengal.

The RSA will start constructing a new spaceport called the Vostochny Cosmodrome. The construction will take place over the first half of the next decade. Major advances will be made in Russian space crafts. Depending on whether or not the commercial space crafts can replace supply to the ISS, the Soyuz will finally be phased out. This will be replaced by the Kliper spacecraft. This re-usable spacecraft will be smaller than the shuttles, but will have a crew capacity of six. The modular approach of the Soyuz will be retained on the Kliper, but the Kliper re-entry module will have wings that will allow for a smoother re-entry than the Soyuz. The Kliper project is expected to be the



The ISRO SCRE capsule post its success

primary access to the ISS in the first half of the next decade.

NASA's budget has been severely curtailed by the heavy expenditure on war by the US. The slowing economy isn't helping too much either. While initial plans post the Apollo missions were on establishing a space habitat or a moon colony by 2000, almost a decade later we are looking at what is essentially a setback with the retirement of the shuttles. The next step by NASA will be to revisit its roots and go back to the drawing board for a new spacecraft. The Crew Exploration Vehicle (CEV) will be manufactured by Lockheed-Martin, and will be based on the earlier Apollo lunar modules, instead of the space shuttle. This craft, widely dubbed as "Orion" will be tested in the first half of the next decade. The crew capacity for the craft will be six, which seems to be a standard maintained by space agencies around the world. Orion will be made up of a crew module, a service module, a docking module and launch abort system. Both the Kliper and the Orion will have a failsafe crew safety in case of a problem during launch. The first real mission of the Orion spacecraft is expected to be in 2014, to serve the ISS.

2015-2030

In these crucial years, the research and development teams from all around the world will focus on permanent and independent human existence in space. This would mean a long-term presence in space, with each unit being self-sustaining. Astronauts, cosmonauts and their variants around the world will be training for longer missions. The CNSA space stations should be in good order by then. Space tourism is also expected to make a comeback after a five-year hiatus, especially if the Russian Kliper missions stay on track.

The ESA is expected to use a modified version of its Automated Transfer Vehicle (ATV) to launch humans beyond low Earth orbit using the Ariane 5 as the launch vehicle. The ATVs are currently being used to re-supply the ISS. A modified version would include life support systems, and at least one module shielded for re-entry. The first launch of this vehicle is expected to take place some time in 2018. Around this time, Korea, Iran and Malaysia are three other countries that will be thinking of sending humans into low-earth orbit. Korea will be depending heavily on China for the necessary technology.

NASA in 2019 will send its Altair spacecraft to the moon for a sortie mission. This will research the essentials for a future lunar base. The Altair spacecraft is a modified version of the Apollo lander, with the biggest difference being in the kind of fuel used. The Altair spacecraft will use cryogenic fuel, which is fuel that requires extremely cold temperatures to be stored in a liquid form.

Some time between 2018 and 2020, the ESA, ISRO and the CNSA all plan to place humans on the moon. The ESA has very clear plans of starting construction on a permanent moon base in 2023, with permanent habitation from 2025. NASA's

